SULIT

FACULTY OF COMPUTING
UTM Johor Bahru

UNIVERSITI TEKNOLOGI MALAYSIA
MID-TERM EXAMINATION SEMESTER 2 2024/2025

SUBJECT CODE	**:**	**SECJ 2253**
SUBJECT NAME	**:**	**REQUIREMENTS ENGINEERING & SOFTWARE MODELING**
SECTION	**:**	**01, 02, 03, 04**
TIME	**:**	**8.00 PM – 10.00 PM**
DATE/DAY	**:**	**02 MAY 2025, FRIDAY**
VENUES	**:**	**BK 1, BK 2 N28 FC**

INSTRUCTIONS:

SECTION A :	**10 MULTIPLE CHOICE QUESTIONS**	**(20 MARKS)**
SECTION B :	**5 STRUCTURED QUESTIONS**	**(80 MARKS)**
TOTAL		**(100 MARKS)**

(Write Your Lecturer Name And Section In Your Answer Booklet)

Name	
Matric No.	
Year / Course	
Section	
Lecturer Name	

This question paper consists of **TEN (10)** printed pages excluding this page.

SECTION A: MULTIPLE CHOICE QUESTION [20 MARKS]

Choose the **BEST** answer for the question. Write the answer in the given booklet.

1. Which of the following **BEST** illustrates a functional requirement for a library management system?
   A. The system must be easy to use by librarians.
   B. The system should handle 100 book transactions per hour.
   C. The system should allow users to search for books by title or author.
   D. The system should be available 99.9% of the time.

2. Which of the following is **NOT** related to the requirements engineering activities?
   A. Prioritizing customer satisfactions
   B. Gathering stakeholder needs
   C. Project feasibility study
   D. Managing requirements attributes

3. Which of the following **BEST** describes a ***grey zone*** in system context analysis?
   A. Areas in the system that are high-risk for failure
   B. System features not documented yet
   C. Elements where it is unclear whether they belong to the system or its environment
   D. Legal constraints that affect development

4. Why is defining the system boundary important during the requirements engineering process?
   A. It determines the training needs of users.
   B. It separates internal system components from external users.
   C. It helps distinguish what is part of the system and what belongs to the environment.
   D. It identifies legal and policy constraints on development.

5. ***Orange***© company is planning to release a smartwatch application by mid-2025, aimed at promoting improved mental well-being in society through advanced AI capabilities. A key feature includes: ***"The smartwatch shall offer an intelligent conversational health assistant that generates motivational affirmations or mindfulness tips based on sensor data."***
   According to the Kano model, which category best describes this feature?
   A. Satisfier
   B. Accuracy
   C. Delighter
   D. Dissatisfier

6. Which of the following **BEST** describes a weakness of prototyping as a requirements elicitation technique?
   A. It promotes a thorough examination of existing requirements.
   B. It can be utilized during system validation.
   C. It primarily emphasizes the user interface's "look and feel."
   D. It can support the documentation process.

7. Which of the following would **NOT** be a relevant source of requirements when developing an investment resource management system for a local Malaysian fintech company?
   A. Applicable Bank Negara Malaysia cybersecurity laws or regulation
   B. Procedures for the export of drugs
   C. Stakeholder needs from the Securities Commission Malaysia (SC)
   D. Business processes within the financial technology industry

8. Which of the following represents the greatest potential risk when using field observation as a technique during requirements elicitation?
   A. It only focuses on performance-related factors.
   B. It requires the observer to immerse themselves in the real operational context.
   C. It may overlook user motivation and direct feedback.
   D. It aims to capture both basic and detailed requirements.

9. Which of the following is the sequence of business process workflow modeling?
   A. Understand As-Is Process → Frame the Process → Develop Use Cases → Design To-Be Process
   B. Frame the Process → Understand As-Is Process → Design To-Be Process → Develop Use Cases
   C. Frame the Process → Design To-Be Process → Understand As-Is Process → Develop Use Cases
   D. Understand As-Is Process → Design To-Be Process → Develop Use Cases → Frame the Process

10. Which of the following statements is **NOT** relevant to a goal model?
    A. It can be documented in natural language.
    B. It can be represented as a hierarchical organization diagram.
    C. It reflects the personal intentions of the requirements engineer.
    D. It features decomposition relationships between broader and more specific goals.

SECTION B: STRUCTURED QUESTION [80 MARKS]

There are **FIVE (5)** questions. The marks for each part of the question are as indicated. Write the answer in the given booklet.

Question 1 [13 marks]

Hopeful Hearts Foundation (HHF) has engaged Maju Software House—where you serve as requirements engineer—to build a Volunteer Orphan Management (VOM) System. HHF partners with non-government organization (NGOs), donors and volunteers to deliver food, shelter, clothing, healthcare and education to orphans, and now seeks to streamline its welfare and volunteer coordination.

The proposed VOM system must be capable of storing and updating orphan and volunteer information, tracking donations and sponsorships, and managing volunteer activity records. The system should integrate with third-party platforms or services that support data analysis, personalized engagement campaigns, and volunteer recognition or reward programs. Additionally, the system must support integration with financial institutions that provide secure online donation gateways, enhancing the donation and sponsorship experience for contributors.

The VOM system should include comprehensive orphan care and volunteer management features, enabling staff to monitor individual orphan progress, track volunteer hours and activities, and receive alerts for required actions such as medical check-ups, schooling milestones, or follow-up visits. The system must ensure accurate recording and tracking of orphans' information, volunteer records, sponsorships, and donations. This is essential to maintain data integrity, compliance, and trustworthiness in dealings with stakeholders. The system should be user-friendly and intuitive, enabling staff members and volunteers to navigate and operate it efficiently with minimal training. Development of the system must adhere to strict budgetary limitations and should be completed within a defined timeframe to meet operational requirements. Furthermore, it must be compatible with the existing hardware and software infrastructure of the foundation to ensure a smooth transition and minimal disruption during implementation.

Based on the above VOM system scenario, answer the following questions (a) until (c):

(a) Specify SIX (6) requirements for the VOM system based on the types; namely : functional, constraint and quality (refer **Table 1**). (6 marks)

Table 1

Types of Requirements	**Requirements Statement**
Functional (FR1)	
Functional (FR2)	
Constraint Requirement (CR1)	
Constraint Requirement (CR2)	
Quality Requirement (QR1)	
Quality Requirement (QR2)	

(b) Describe TWO (2) key responsibility of a requirements engineer in defining and analyzing requirements during the development of the proposed VOM system. (4 marks)

(c) How can effective Requirements Engineering activities support the successful development of the VOM system? (3 marks)

Question 2 **[13 marks]**

Kuala Kubu (KK) University is planning to develop a new Hostel Management (HM) system to streamline and automate core hostel operations, including room allocation, student check-in/check-out, and maintenance request management. The system will utilize a centralized student occupancy database, enabling wardens to monitor room availability, assign rooms based on eligibility, manage student preferences, and track room condition status.

The HM system must interact with several key users, including students, wardens, finance officers, and management staff. These users will rely on the system to access accurate, real-time information and perform role-specific functions such as lodging complaints, approving maintenance jobs, or verifying student eligibility.

In terms of system integration, the university is evaluating whether to include a hostel fee payment module in this initial phase. If implemented, it must seamlessly integrate with the university's existing Finance System to ensure accurate billing and timely payment tracking. Additionally, the HM system should ensure compliance with the Personal Data Protection Act (PDPA) by implementing strong access control, consent management, and secure data handling practices to safeguard student information.

The HM system is also expected to interface with existing Student Information Systems (SIS) to retrieve verified student records for room allocation and eligibility checks. Integration with Access Control Systems (e.g., RFID-based door systems) may be needed to track student movement in/out of hostel buildings. If outsourced, a Maintenance Management System must also be linked to forward student requests and receive job status updates.

Based on the above HM system of the KK University context, answer the following questions (a) until (c):

(a) Define the term of interfaces in the HM system context. (2 marks)

(b) Identify TWO (2) stakeholders and TWO (2) systems in operation in the HM system context. (5 marks)

(c) Examine TWO (2) potential grey areas that may arise during the development of the HM system at KK University. Your analysis should relate each grey area to the relevant system context objects (SCOs) that could be impacted by the proposed enhancements (refer **Table 2**). (6 marks)

Table 2

Grey Zone Issue (1 mark)	**Related System Context Object (SCO) (1 mark)**	**Justification (1 mark)**

Question 3 **[34 marks]**

Universiti Teknologi Malaysia is modernizing its Industrial Training/Internship Management System (ITS) to better serve a diverse user base that includes students, faculty advisors, faculty coordinators, and industry partners. The new ITS must seamlessly integrate with the existing MyAIMS system—which provides access to student profiles and course registration—to ensure the integrity of up-to-date academic records and enrolment information while incorporating robust authentication and authorization mechanisms.

The university's administration has launched a comprehensive requirements elicitation project to guarantee that the revamped ITS addresses all stakeholder needs. The project team is charged with identifying, categorizing, and prioritizing system requirements using a range of elicitation techniques and models. As part of this process, current internship policy manuals, academic standards, and established guidelines will be thoroughly reviewed, and the performance and documentation of the legacy system will be carefully analysed.

During individual elicitation sessions *with faculty coordinators and industry partners, participants emphasized that the new ITS should feature a user-friendly interface to streamline application submissions, provide effective feedback, and monitor progress efficiently. They also highlighted the necessity for comprehensive dashboards and real-time reporting tools that offer valuable insights into internship metrics for both students and administrators.*
In addition to these core features, the project team is exploring innovative enhancements such as AI-powered recommendations that can match students with the most appropriate internship opportunities based on their skills, academic performance, and career aspirations. This forward-thinking approach aims to deliver an intelligent and adaptive system that not only meets current operational demands but is also primed for future advancements.

Based on the above new UTM-ITS system scenario, answer the following questions (a) until (e):

(a) Identify **THREE (3)** type of requirements sources that can be consulted during elicitation process, and provide a corresponding example for each (refer **Table 3**). (6 marks)

Table 3

Sources of Requirements	**Example**

(b) Propose **FOUR (4)** stakeholders satisfaction for the new ITS system, each aligned to Kano model factor (refer **Table 4**). (8 marks)

Table 4

Kano Factor	**Stakeholders Satisfaction**
Basic (Dissatisfier)	
Performance (Satisfier)	
Performance (Satisfier)	
Excitement (Delighter)	

(c) Which elicitation technique is being referenced by the phrase ***"During individual elicitation sessions…"*** in the context of the new UTM-ITS system? Explain how this technique can be applied, describe the steps involved in its implementation, and identify the specific Kano Model factor that would be most relevant to observe. (6 marks)

(d) Propose four distinct elicitation techniques—each from a different category than the one used in part (c)—that the project team could effectively employ. For each proposed technique, justify why it is appropriate and identify the relevant stakeholders who should be involved. Refer to **Table 5** to guide your answer. (8 marks)

Table 5

Category	**Requirements Elicitation Technique & Justification**
Observation	
Document centric	
Creativity	
Supporting	

(e) Identify the most suitable prototyping technique for the scenarios in **Table 6.** Justify the reason. (6 marks)

Table 6

Scenario	Prototyping technique (1mark)	Reason (2 marks)
i. *Quickly develop a prototype of the student dashboard interface to gather immediate feedback on design and usability before a full-scale implementation.*		
ii. *Implement a complex integration module with the MyAIMS system to manage real-time academic data exchange.*		

Question 4 **[5 marks]**

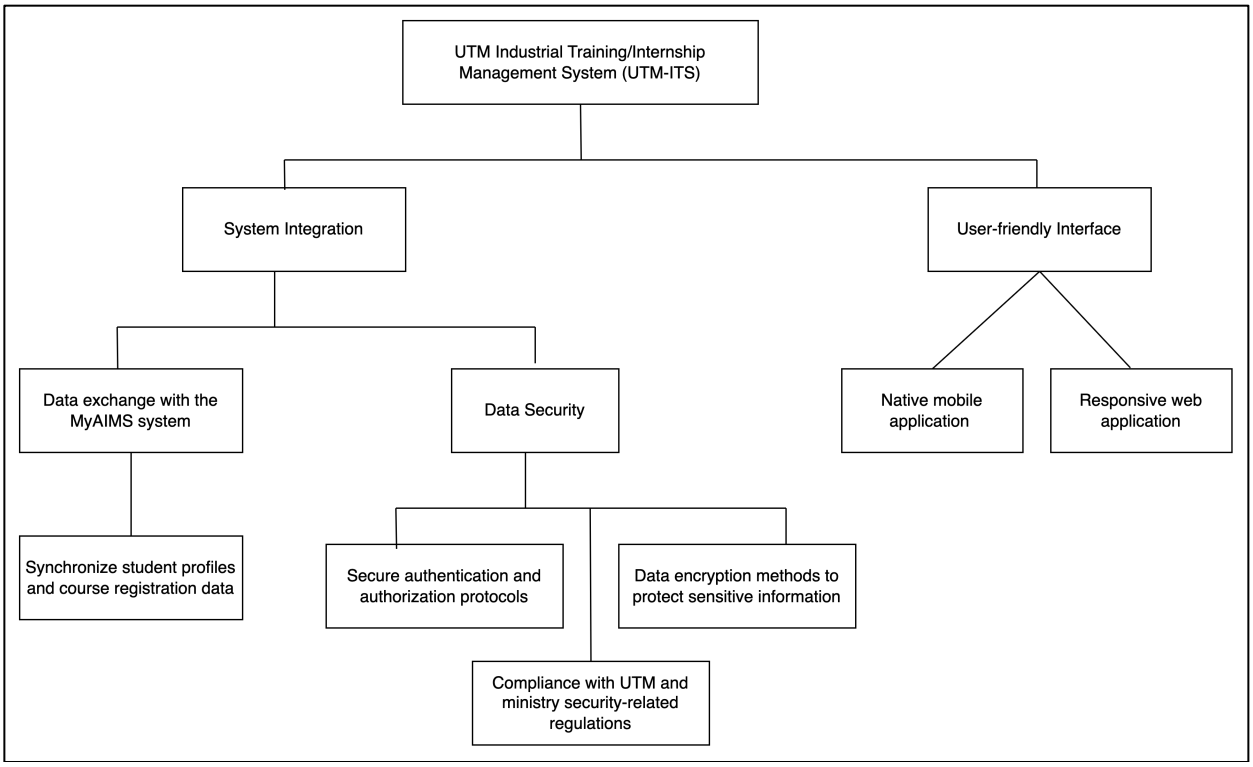


Figure 1: Initial Goal Model of To-Be UTM-ITS system

(a) Describe the goal model for To-Be UTM-ITS system and its decomposition in Figure 1. (3 marks)

(b) Suggest ONE (1) enhancement for the initial goal model of To-Be UTM-ITS system. Justify the enhancement. (2 marks)

Question 5 **[13 marks]**

Figure 2 illustrates the existing (As-Is) process for scheduling an appointment with a therapist at MindCare Special Needs Centre. Customer will first sends a message to request an appointment. The admin then checks the appointment schedule manually and sends the appointment details to customer. If the customer decides to cancel, they may send a cancellation message at any point before the appointment. Else, if the appointment is accepted by customer, the admin requests the customer to complete a form with necessary information. Once the customer form is submitted, the admin manages the appointment by recording and organizing the schedule. The head of department (HoD) later validates the finalized schedule to ensure it complies with departmental policies and availability. After validation by HoD, the admin sends an email reminder to the customer, who then receives a notification of the confirmed appointment.

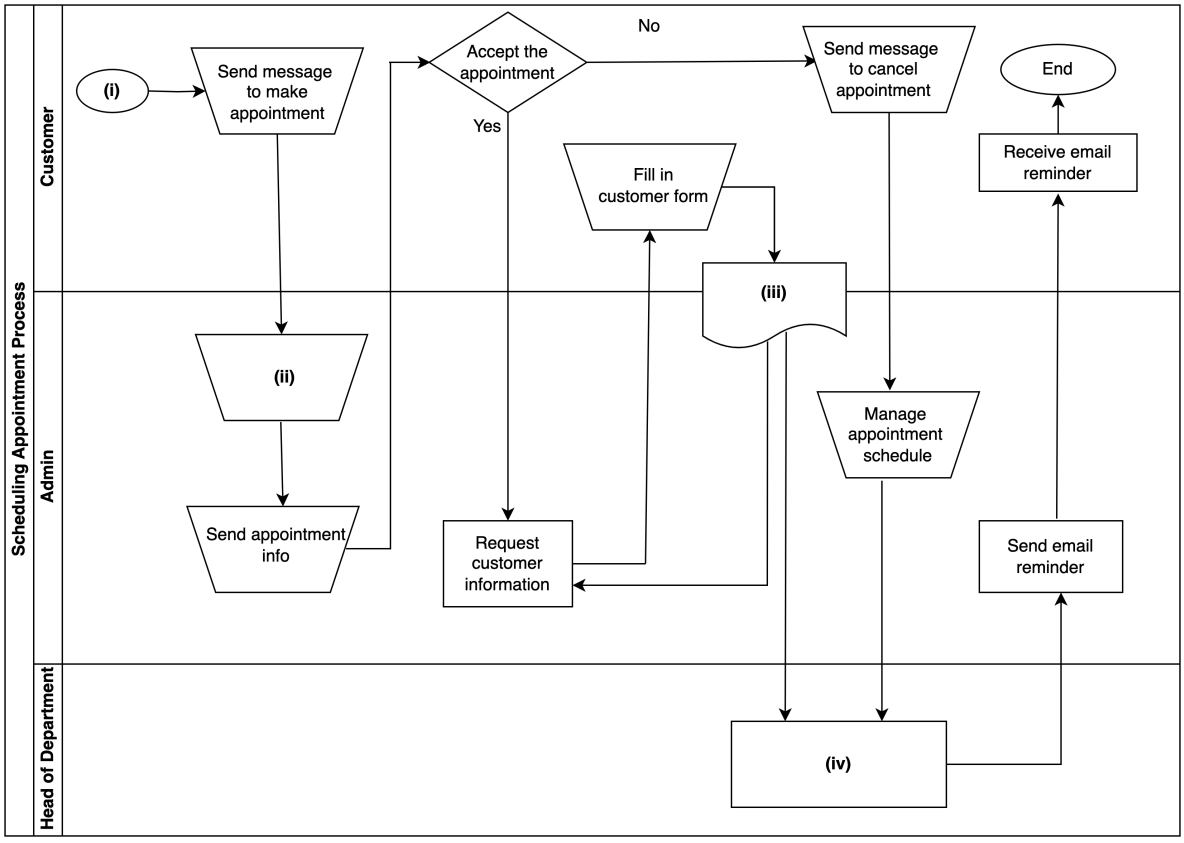


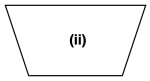
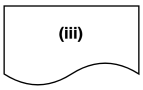
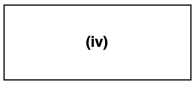
Figure 2: Swimlane Diagram of Scheduling Appointment Process

Given the above scenario for the scheduling appointment process at MindCare Special Needs Centre, answer the following (a) and (b) questions:

(a) Describe **TWO (2)** main purposes of using a swimlane diagram to visualize a business process workflow. (5 marks)

(b) Analyze the workflow of scheduling appointment process in the Figure 2. Complete the details of the swimlane diagram elements (refer **Table 7**). (8 marks)

Table 7

Element	Symbol Meaning (1 mark)	Element Details (1mark)
		
		
		
		

- END OF QUESTIONS -